prolfqua - R-package for Proteomics Label Free Quantification using Linear Models

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Introduction

We use `prolfqua` to develop highly customizable, visually appealing, and interactive data analysis reports in pdf or HTML format for quantification experiments. We use `prolfqua` to visualize and model simple experimental designs with a single explanatory variable and complex experiments with multiple factors. The `prolfqua` package integrates essential steps of the data analysis workflow: quality control, data normalization, protein aggregation, sample size estimation, modeling, and hypothesis testing. We further use `prolfqua` to benchmark data acquisition, data preprocessing or data modeling methods. We developed and improved the package by applying the "Eating your own dog food" principle, making it easy to use.

Implementation & Methods

We store all the data needed for analysis in a single data frame in a tidy table, i.e., every column is a variable, every row is an observation, every cell is a single value. Using an **R6** configuration object, we specify what variable is in which column, making it easy to integrate new inputs in `prolfqua` if provided in tidy tables. For example, to visualize tidy *Spectronaut*, or *Skyline* outputs, or data in *MSStats* format, only a few lines of code to update the `prolfqua` configuration are needed. For popular software like *MaxQuant* or *MSFragger*, which stores the same variable (e.g., intensity) in multiple columns, one for each sample, we implemented methods that transform the data into tidy tables. Relying on the tidy data table enabled us to easily interface with many data manipulation, visualization, and modeling methods implemented in base **R** and the tidyverse. We use **R6** classes to structure the functionality of the package (see Figure 1 and Figure 2). **R6** classes are well supported (e.g. auto-completion) and easy to use (OOP).

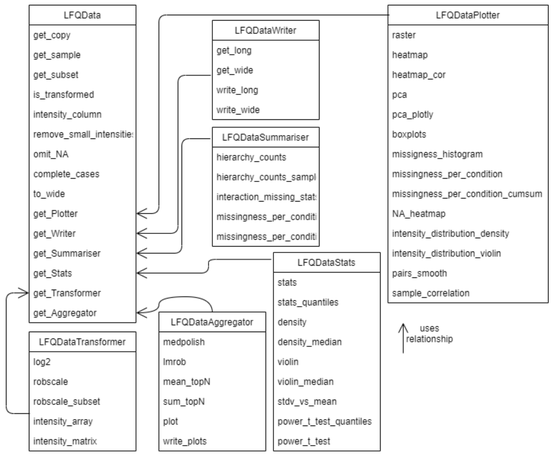

Figure 1: Class Diagram for LFQData and related objects. The LFQData_Plotter class uses the LFQData class to implement methods for plotting.

R linear model and linear mixed effect models allow modeling parallel designs, repeated measurements, factorial designs, and many more. **R**'s formula interface for linear models is flexible, widely used, and well documented. This makes it easy to reproduce an analysis performed with `prolfqua` in other statistical programming languages. We implemented features specific to high throughput experiments, such as the experimental Bayes variance and p-value moderation, which utilizes the parallel structure of the protein measurements and the analysis (Ritchie et al. 2015). We also compute probabilities of differential protein regulation based on peptide level models (Suomi and Elo 2017). We use R6 to model the statistical modeling functionality in `proflqua` (see Figure 2).

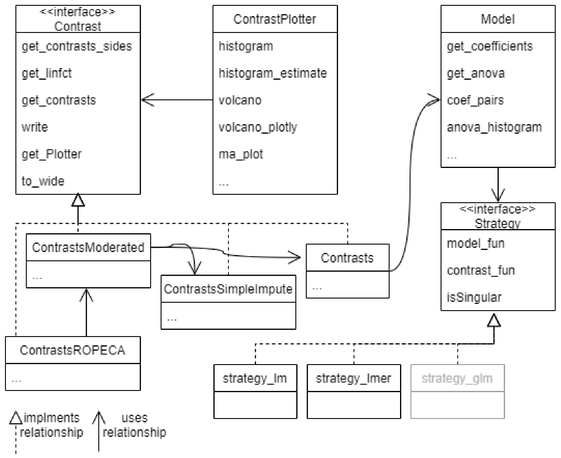

Figure 2: UML diagram of modeling and contrast related classes. Different strategies e.g. *lm*, *lmer*, and *glm* can be used to fit the models using a model builder (see code). All classes computing contrasts implement the *Contrast* interface.

Example Analysis

The IonStar (Shen et al. 2018) benchmark dataset contains *H. sapiens* proteins with constant concentrations and *E. coli* proteins with varying concentrations. We know that for *H. sapiens* proteins the difference β between two dilutions should be $\beta = 0$ while for *E. coli* proteins, we know that the difference between dilutions should be $\beta \neq 0$. For benchmarking we used the contrasts resulting in small fold-changes $\beta = 1.2, 1.25, 1.3(3), 1.5$. Peptide intensities are $\log_2$ transformed and robust z-score scaled but then rescaled to preserves the original range of the data. Protein intensities are estimated using Tukey's median polish (see code example and Figure 3).

```
d <- prolfqua::data_ionstar$filtered()
lfqd <- LFQData$new(d$data, d$config)
t <- lfqd$get_Transformer()
lfqd <-  t$log2()$robscale()$lfq
pl <- lfqd$get_Plotter()
p_1 <- pl$intensity_distribution_density() +
  labs(tag = "A") + theme(legend.position = "none")
agr <- lfqd$get_Aggregator()
lfqp <- agr$medpolish()
p_2 <- agr$plot()$plots[[1]] + labs(tag = "B")
p_3 <- lfqp$get_Stats()$violin() + labs(tag = "C")
pl <- lfqp$get_Plotter()
p_4 <- pl$boxplots()$boxplot[[1]] + labs(tag = "D")
ggpubr::ggarrange(p_1, p_2, p_3, p_4)
```

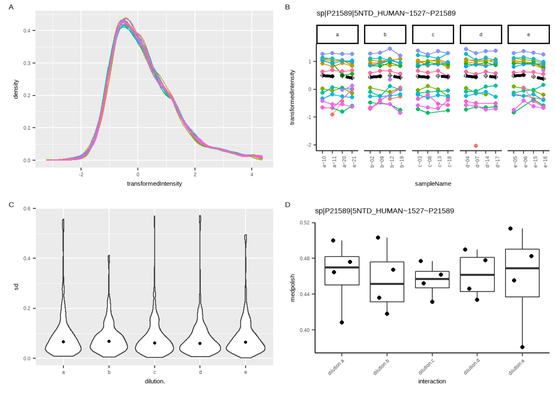

Figure 3: Panel A - Peptide intensity distributions for 20 samples, Panel B - Peptide intensities for protein *5NTD* and protein intensity estimate (black dashed line), Panel C - distribution of protein standard deviations for all dilutions, Panel D - Distribution of protein intensities for protein *5NTD*.

```
contrasts <- c(
  "dilution_(9/7.5)_1.2" =   "dilution.e - dilution.d",
  "dilution_(7.5/6)_1.25" =   "dilution.d - dilution.c",
     "dilution_(6/4.5)_1.3(3)"   =        "dilution.c   -
        dilution.b",
  "dilution_(4.5/3)_1.5" =   "dilution.b - dilution.a"
)
# fit model
lmmodel <- paste(lfqp$intensity_column(()," ~ dilution.")
modelFunction <- strategy_lm( lmmodel, model_name = "lm")
models <- build_model(lfqp, modelFunction)
p1 <-  models$anova_histogram()
# compute contrasts from linear model and with imputation
contr <- Contrasts$new(models, contrasts) %>%
  ContrastsModerated$new()
conI <- ContrastsSimpleImpute$new( lfqp, contrasts) %>%
  ContrastsModerated$new()
tmp <- addContrastResults(contr, conI)
# visualize results
pl <- tmp$merged$get_Plotter()
p1 <- pl$histogram()$p.value
p2 <- pl$volcano()$FDR +
   theme(legend.position = "bottom")
gridExtra::grid.arrange(p1, p2, ncol = 2)
```

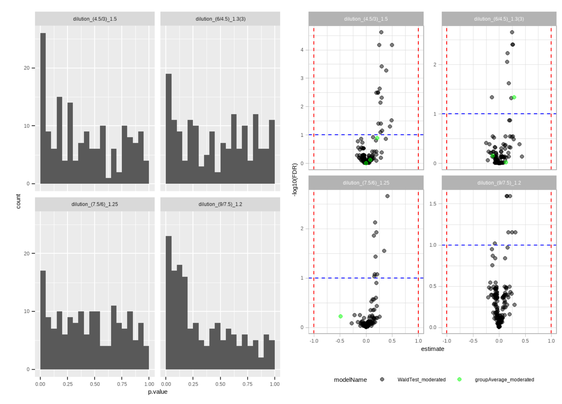

Figure 4: Left panel - histogram of p-values. Right panel - volcano plots.

Benchmarking

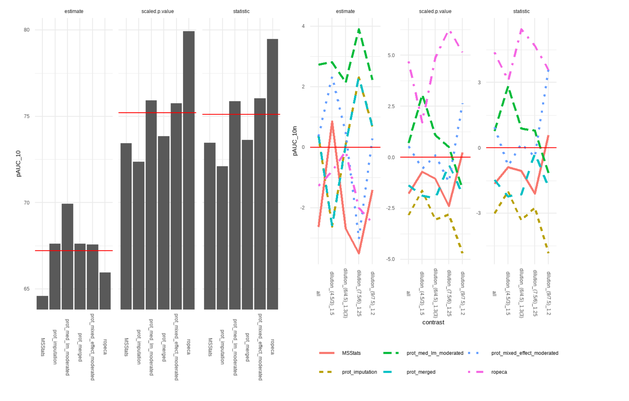

Figure 5: Left panel - Partial area under the ROC curve at 10% FPR for all contrasts. Red line average area under the curve. Left panel - Difference to mean partial area under the ROC curve for various models, at 10% FPR, by fold change.

The Benchmark functionality of `prolfqua` includes ROC curves and computes partial areas under those curves (pAUC) and other scores (`?Benchmark`). We use it to study how well linear, mixed effect models or p-value moderation model quantitative mass spectrometric high throughput experiments and compare it with results produced by MSstats (Choi 2014). Figure 5

A relevant parameter is the number of proteins for which we estimated the contrasts (see Figure 6). It indicates how robust the models are in the presence of missing data. Also important is if the FDR produced is an unbiased estimate of the false discovery proportion (FDP) (see Figure 6 right panel).

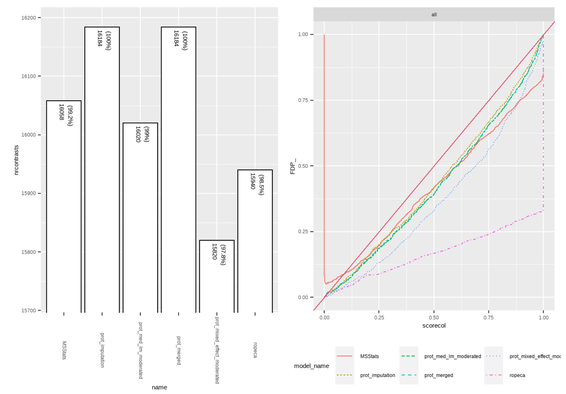

Figure 6: Left panel - Number and percentage of estimated contrasts by modeling method. Right panel - Compare FDR estimate with false discovery proportion (FDP).

Conclusions

`prolfqua` is an easy-to-use package to analyze quantitative mass spectrometric data, report results, and benchmark MS software and statistical methods. We provide more details at the website https://wolski.github.io/prolfqua/. To install:

```
install.packages('remotes')
remotes::install_gitlab("wolski/prolfquaData",
                        host="gitlab.bfabric.org")
remotes::install_github('wolski/prolfqua',
                        build_vignettes = TRUE)
```


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References

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Ritchie, Matthew E, Belinda Phipson, Di Wu, Yifang Hu, Charity W Law, Wei Shi, and Gordon K Smyth. 2015. "Limma Powers Differential Expression Analyses for RNA-Sequencing and Microarray Studies." *Nucleic Acids Research* 43 (7): e47–47.

Shen, Xiaomeng, Shichen Shen, Jun Li, Qiang Hu, Lei Nie, Chengjian Tu, Xue Wang, et al. 2018. "IonStar Enables High-Precision, Low-Missing-Data Proteomics Quantification in Large Biological Cohorts." *Proceedings of the National Academy of Sciences* 115 (21): E4767–76.

Suomi, Tomi, and Laura L Elo. 2017. "Enhanced Differential Expression Statistics for Data-Independent Acquisition Proteomics." *Scientific Reports* 7: 5869.